

Reg No.: \_\_\_\_\_

Name: \_\_\_\_\_

**APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY**

B.Tech Degree S6 (R, S) / S6 (PT) (R) Examination June 2023 (2019 Scheme)

**Course Code: EET302****Course Name: LINEAR CONTROL SYSTEMS**

Max. Marks: 100

Duration: 3 Hours

**PART A***Answer all questions, each carries 3 marks.*

Marks

- |    |                                                                                                                               |     |
|----|-------------------------------------------------------------------------------------------------------------------------------|-----|
| 1  | How does closed-loop system differ from open-loop system in its function?                                                     | (3) |
| 2  | Give a comparison between lead and lag compensator.                                                                           | (3) |
| 3  | Name the time domain performance specification for first-order systems and define it.                                         | (3) |
| 4  | What pole locations characterize (1) the underdamped system, (2) the overdamped system, and (3) the critically damped system? | (3) |
| 5  | What are two ways to find where the root locus crosses the imaginary axis?                                                    | (3) |
| 6  | Write the magnitude and angle conditions for the root locus of positive feedback systems.                                     | (3) |
| 7  | Discuss how to obtain the frequency response of a system analytically.                                                        | (3) |
| 8  | Define gain margin and phase margin of a system.                                                                              | (3) |
| 9  | State the Nyquist criterion. What does the Nyquist criterion tell us?                                                         | (3) |
| 10 | Explain the importance of Nichols chart.                                                                                      | (3) |

**PART B***Answer one full question from each module, each carries 14 marks.***Module I**

- |    |                                                                                                                                                                  |     |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 11 | a) Obtain the transfer function of lead compensators using R-C circuit components and bring out the characteristics of lead compensators.                        | (7) |
|    | b) An automobile driver uses a control system to maintain the speed of the car at a prescribed level. Sketch a block diagram to illustrate this feedback system. | (7) |

**OR**

- |    |                                                                                                                            |     |
|----|----------------------------------------------------------------------------------------------------------------------------|-----|
| 12 | a) Derive the transfer function for the armature-controlled DC motor using the block diagram representation of the system. | (6) |
|    | b) List the control applications of (i) Tacho Generator and (ii) Synchro                                                   | (8) |

**Module II**

- 13 a) A unity negative feedback control system has the loop transfer function (8)

$G(s) = \frac{K}{s(s+\sqrt{2K})}$ . (i) Determine the percent overshoot and settling time (using a 2% settling criterion) due to a unit step input. (ii) For what range of K is the settling time,  $T_s \leq 1$  sec?

- b) A system with unity feedback is having an open loop transfer function (6)

$G(s) = \frac{20}{s^2+14s+50}$ , determine the steady-state error for a step and a ramp input.

**OR**

- 14 a) The system shown in Figure 1 has a unit step input. Find the output response as a function of time. Assume the system is underdamped. (7)

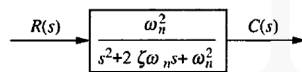


Figure 1

- b) A system has a characteristic equation  $s^5 + s^4 + 2s^3 + s^2 + s + K = 0$ . Determine the range of K for stability. (7)

**Module III**

- 15 a) A unity feedback system has the loop transfer function  $G(s) = \frac{K(s+2)}{s(s+1)}$ . Sketch the root locus and find the gain when the closed loop poles of the system are located at  $s_{1,2} = -2 \pm \sqrt{2}j$ . (10)

the root locus and find the gain when the closed loop poles of the system are located at  $s_{1,2} = -2 \pm \sqrt{2}j$ .

- b) Explain the effect of adding poles and zeros to the nature of root locus. (4)

**OR**

- 16 a) Sketch the root locus for the unity feedback system with open loop transfer function  $G(s) = \frac{K(s^2+1)}{s^2}$ . Comment on the stability of closed loop system from root locus. (10)

function  $G(s) = \frac{K(s^2+1)}{s^2}$ . Comment on the stability of closed loop system from root locus.

- b) Explain the steps involved in the design of lag compensator using root locus method. (4)

**Module IV**

- 17 a) Sketch Bode Plot for the system with open loop transfer function (10)

$G(s) = \frac{100}{s(s+10)^2}$ . From Bode plot, determine GM and PM of the system and assess the stability of the system.

- b) Describe the change in the magnitude and phase plot of the system if time delay is added to the system. (4)

OR

- 18 a) Sketch Polar Plot for the following Open Loop Transfer Function given below (10)

$$G(s)H(s) = \frac{10e^{-0.2s}}{s+1}.$$

- b) Define resonant frequency, resonant peak, Bandwidth and cut off rate for a standard second order system. (4)

**Module V**

- 19 a) Draw Nyquist plot for the system whose open loop transfer function is (10)

$$G(s) = \frac{6}{s(1+s)(2+s)}.$$

Determine the number of right-hand side poles of the system and assess the stability.

- b) Explain the design constraints on the selection of corner frequencies of lag compensator. (4)

OR

- 20 a) Compare the Nyquist plots of (i)  $G(s) = \frac{K}{sT_1+1}$ , (ii)  $G(s) = \frac{K}{s(sT_1+1)}$ , (9)

$$(iii) G(s) = \frac{K}{s^2(sT_1+1)}$$

and comment about stability of the system.

- b) Explain the steps involved in the design of lead compensator using bode plot. (5)

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